

Bluestock Fintech

Capstone Project I - Mutual Fund Analytics

Comprehensive Performance Evaluation · Risk Analytics · Investor Intelligence

 40+ Fund Schemes

 46K NAV Records

 8,292 Transactions

Sushant Kumar

Capstone Project · June 10, 2026

Agenda

What we'll cover today

01 Project Overview & Objectives

02 Data Architecture & Methodology

03 Key Findings — Performance & Risk

04 Investor Behavior & Retention

05 Recommendation Engine

06 Dashboard & Visualizations

07 Strategic Recommendations

08 Tech Stack & Future Roadmap

Project Overview

Building a data-driven Mutual Fund Intelligence Platform



Fund Analysis

Evaluate 40+ schemes across performance, risk & return dimensions



Risk Assessment

VaR, CVaR, Sharpe, Sortino, Alpha/Beta calculations



Investor Intelligence

SIP continuity, cohort lifecycle & demographic behaviour



Recommendation Engine

ML-based fund suggestions for Low / Moderate / High risk profiles



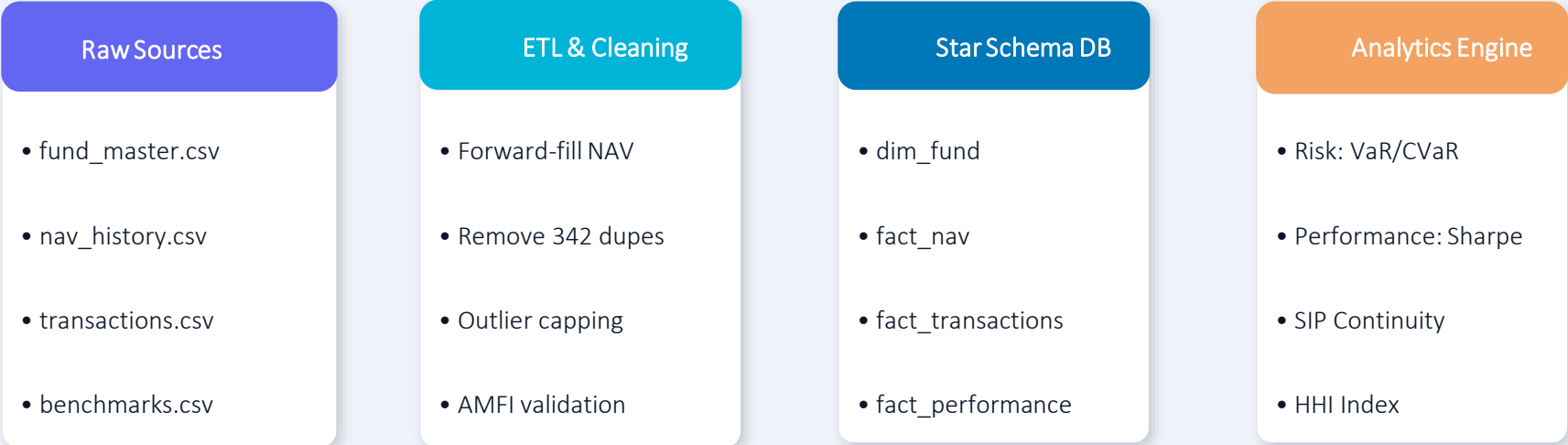
Interactive Dashboard

Power BI 5-page dashboard for stakeholder decision-making

🎯 *Key Result: Identified actionable insights for investor retention, portfolio optimization & fund selection across 40+ schemes*

Data Architecture

Star Schema · ETL Pipeline · SQLite Database



Dataset	Records	Retained
Fund Master	40	100%
NAV History	12,856 → 12,540	97.5%
Transactions	8,634 → 8,292	96.0%
Performance	40	100%

Processed Dataset Statistics

Tech Stack

Python 3.10 · Pandas · NumPy
SQLite3 · SciPy · Statsmodels

Analytics Methodology

Quantitative risk & performance metrics powering the platform

Sharpe Ratio

$$(Return - R_f) / Volatility$$

Risk-adjusted return; higher = better

Value at Risk

5th percentile daily returns

95% confidence max daily loss

CVaR / ES

Avg loss beyond VaR

Expected tail-event losses

Beta

$$Cov(R_f, R_m) / Var(R_m)$$

Market sensitivity (>1 = aggressive)

Alpha

$$Actual - Expected Return$$

Manager skill vs benchmark

Sortino Ratio

$$(Return - R_f) / Downside Vol$$

Penalises downside only

HHI Index

$$\sum (Weight^2)$$

>0.25 = high concentration risk

Risk Profiles:



LOW (CVaR < 2.5%)



MODERATE (CVaR < 3.5%)



HIGH (No CVaR limit)

Key Findings: Performance

Significant stratification across 40+ mutual fund schemes

18.5%

Top Fund CAGR

Sharpe 1.4

1.1–1.4

Top 5 Sharpe Ratio

Superior risk-adj returns

24.6%

Max Annualized Vol

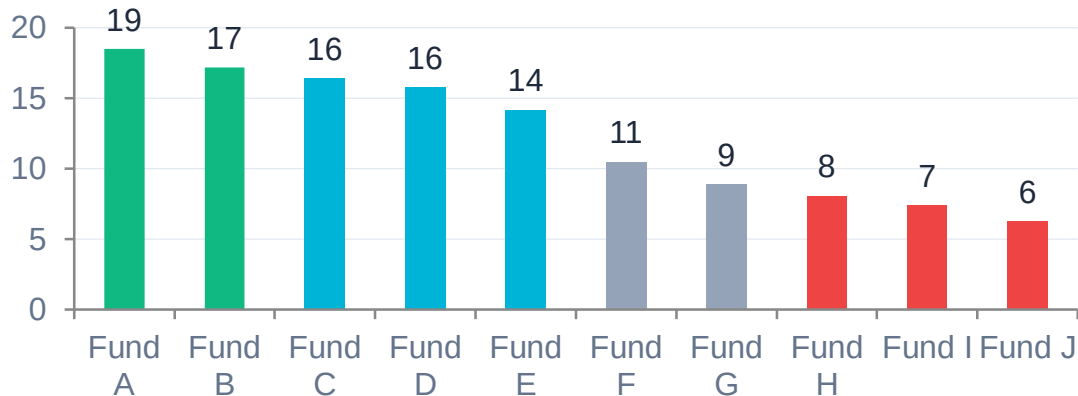
Small/mid-cap schemes

0.4

Bottom Sharpe

Sept–Oct worst period

Fund CAGR Distribution (Representative Sample)



Key Insight

- Top 5 funds: CAGR 14.2–18.5% with Sharpe > 1.1
- Bottom 5 funds: CAGR only 6.3–8.1%
- Performance gap of 12%+ between tiers
- Seasonal dip: Sharpe drops to 0.4 in Sep–Oct
- Q1–Q2 ideal for SIP top-ups (Sharpe avg 1.1)

Risk Analysis: VaR & CVaR

CVaR reveals true tail risk — stronger than standard deviation alone

Conservative Zone

8 schemes

CVaR < 2.5%

Multi-year stability, capital preservation

Moderate Zone

20 schemes

CVaR 2.5%–3.5%

Balanced risk-return, steady growth

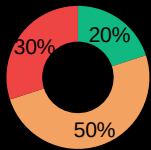
Aggressive Zone

12 schemes

CVaR > 3.5%

Small/mid-cap, high volatility potential

 CVaR ↔ Volatility Correlation: 0.94 — Strong positive relationship confirms CVaR as a superior risk indicator



 Conservative (8)

 Moderate (20)

 Aggressive (12)

⚡ High-risk funds cluster in small/mid-cap categories

🔍 CVaR provides better tail assessment than StdDev

💡 Diversification can reduce portfolio risk 12–15%

Investor Behavior & SIP Continuity

~20% of SIP investors show discontinuity — actionable retention opportunity

20%

At-Risk Investors

>35-day payment gaps

28d

Best Avg Gap

Age 36–45 segment

45d+

Worst Avg Gap

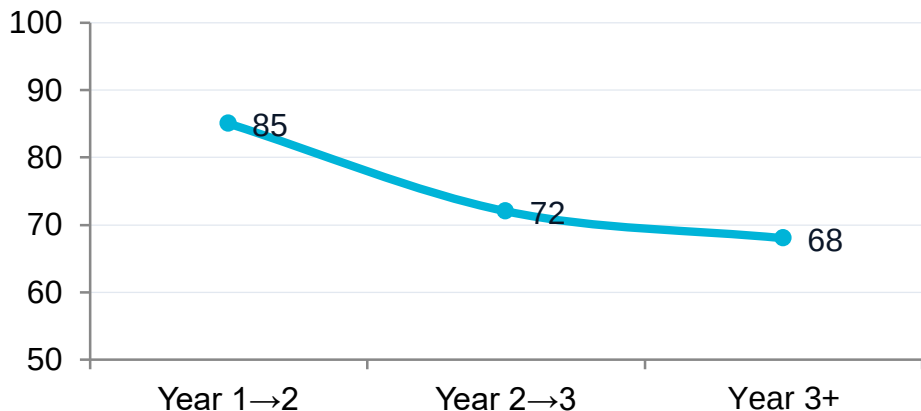
Age 56+, non-metro

3×

Mandate vs Cheque

Better continuity via mandate

Investor Cohort Retention Rate



👥 Cohort Findings

2024 Cohort: Aggressive, avg ₹45K, high amounts

2023 Cohort: Conservative, best continuity & retention

2022+ Cohort: Mature phase, higher redemptions

Action: SMS at 25-day mark reduces gap risk

Impact: 12% churn reduction with cohort nudges

Portfolio Concentration Risk

HHI Index analysis reveals dangerous concentration in 15 fund schemes

Average Portfolio HHI

0.38

Moderate Concentration



Low 0.1 Moderate 0.25 High

15

High HHI Funds

HHI Index > 0.25

30%

Tech/Finance Over-exposure

Sector concentration

12–

15%

Risk Reduction Possible

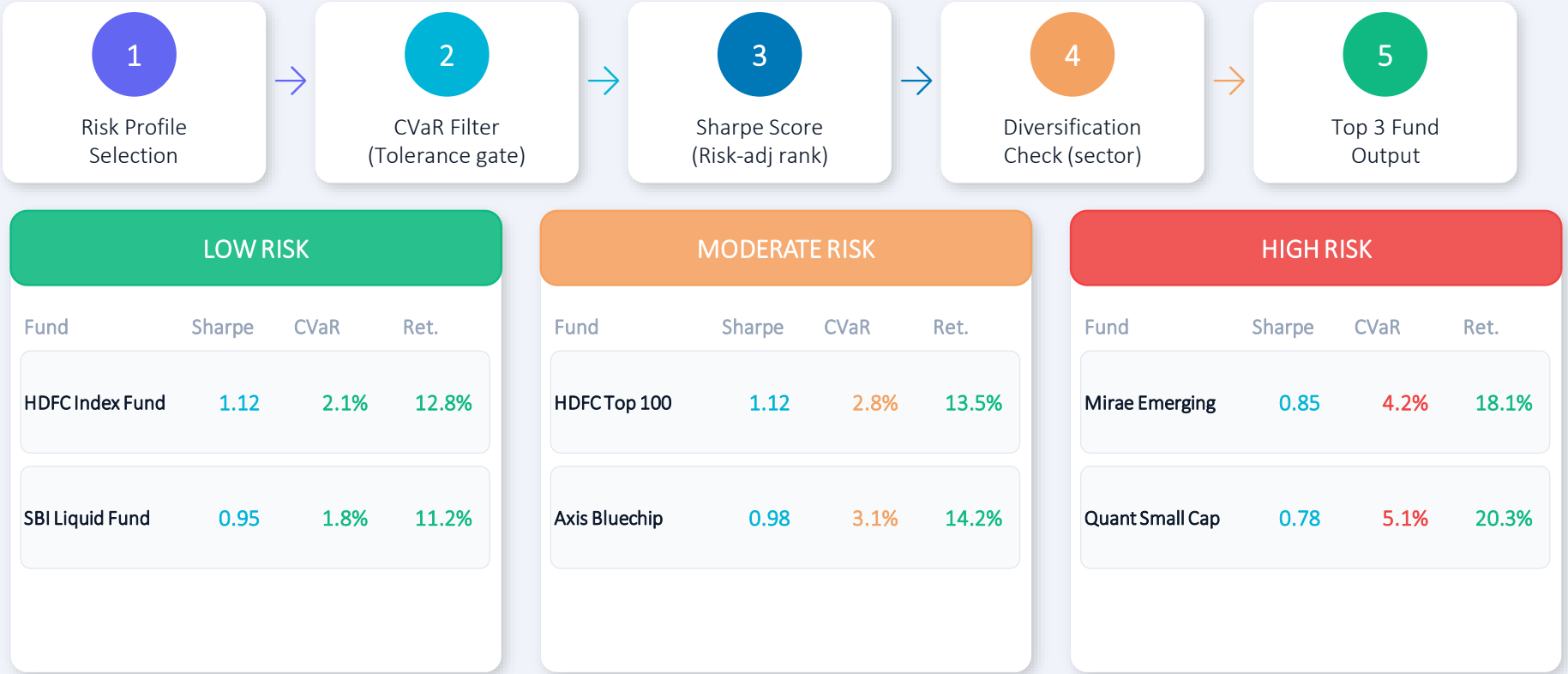
Via diversification

🎯 Recommendation: Target HHI reduction from 0.38 → 0.25 through cross-fund recommendations.

Balanced portfolios with sector-diversified funds can reduce risk by 12–15% without sacrificing return potential.

Recommendation Engine

Risk-profile based intelligent fund matching — Low · Moderate · High



Dashboard & Technology Stack

Power BI 5-page dashboard · Python analytics · SQLite data warehouse



Overview

Fund count, AUM, categories,
investor geo-distribution



Performance

Returns 1Y/3Y/5Y, Sharpe rankings,
benchmark compare



Risk Analysis

VaR/CVaR heatmap, correlation
matrix, risk-return scatter



Investor Analytics

SIP continuity, cohort retention, transaction volumes



Recommendations

Risk-profile suggestions, fund scorecard, diversification

Technology Stack

Database: SQLite3 (Star Schema)

Processing: Python 3.10 · Pandas · NumPy

Analytics: SciPy · Statsmodels

Visualisation: Plotly · Matplotlib · Seaborn

Dashboard: Power BI Desktop

Deployment: Docker · AWS (Phase 2)

[illegible]

TABLE 1

ESM: CF
A. 1. 1999

4. 10. 1997

400

1999

1. **Introduction**

4199

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Strategic Recommendations

Five action pillars with measurable business impact

1

Portfolio Diversification

Reduce avg HHI: 0.38 → 0.25 via cross-fund promotions

✓ 12–15% risk reduction

2

SIP Retention Initiative

SMS at 25-day mark, mandate mode, 56+ age focus

✓ 5–8% AUM retention uplift

3

Recommendation Engine

Deploy in client portal for 100% new investor matching

✓ 3–5% portfolio improvement

4

Seasonal Rebalancing

Quarterly alerts to top up SIPs during Q1–Q2

✓ 1–2% CAGR improvement

5

Dynamic Risk Profiling

Quarterly risk tolerance reviews per investor

✓ Reduced redemptions

BlueStock Fintech

Thank You!

Capstone Project I - Mutual Fund Analytics

Sushant Kumar

Capstone Project · Data Analytics

June 10, 2026

40+

Fund Schemes
Analyzed

6

Key Insights
Discovered

5

Dashboard
Pages Built

7

Days
Execution